

Pregunta 22.7

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Series de tiempo

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Colab

https://colab.research.google.com/drive/1hFbuaQLd3Paba-U-1kWpMUKlo7-MYdOQ?usp=s_haring

Prompt

<https://grok.com/chat/f95f6355-8bbb-4084-90a6-6bc09b3914ef>

Github

<https://github.com/A00573236/Pregunta-22.7>

Interpretations

Stochastic Nature

Both GS3M and GS6M series exhibit stochastic behavior, as they are time series with random fluctuations driven by economic factors. The ARIMA models (order (3,1,2) for both) capture this stochasticity through autoregressive (AR) and moving average (MA) components, indicating that past values and shocks influence future values. The need for differencing ($d=1$) suggests both series follow a stochastic trend, potentially a random walk with drift, which is common in financial time series like Treasury Bill rates.

Stationarity and Unit Roots

- **GS3M:**
 - **ADF Test:** p-value = 0.5140 (> 0.05) fails to reject the null hypothesis of a unit root, suggesting non-stationarity.
 - **KPSS Test:** p-value = 0.0100 (< 0.05) rejects the null of stationarity, confirming non-stationarity.
 - **Interpretation:** GS3M likely has a unit root, indicating a non-stationary stochastic process. The ARIMA model's $d=1$ confirms that first differencing is required to achieve stationarity.
- **GS6M:**
 - **ADF Test:** p-value = 0.0032 (< 0.05) rejects the null of a unit root, suggesting stationarity.
 - **KPSS Test:** p-value = 0.0100 (< 0.05) rejects the null of stationarity, indicating non-stationarity.
 - **Interpretation:** Conflicting results suggest GS6M may be borderline stationary or have a weak unit root. The ARIMA model's $d=1$ indicates that differencing is still applied, likely to ensure robustness in modeling a potentially non-stationary series.

The presence of unit roots in both series (especially GS3M) implies that shocks to these rates have persistent effects, a hallmark of non-stationary stochastic processes. The KPSS warnings about p-values being outside the lookup table suggest strong non-stationarity, reinforcing the need for differencing.

Correlograms

While correlograms are not explicitly plotted, the ARIMA(3,1,2) models imply significant autocorrelation structures:

- **AR(3)**: Indicates that the series have memory up to three lags, suggesting that past rates (up to three periods) influence current rates.
- **MA(2)**: Suggests that random shocks from the past two periods affect the current value, smoothing out short-term fluctuations. A correlogram of the original series would likely show slowly decaying autocorrelations (indicative of non-stationarity), while the differenced series would show autocorrelations dropping to zero quickly, confirming stationarity after differencing.

Cointegration

Cointegration analysis was not explicitly performed, but given that GS3M and GS6M are 3-month and 6-month Treasury Bill rates, they are likely cointegrated. Both series are potentially I(1) (integrated of order 1, as $d=1$ in ARIMA), and economic theory suggests that short-term and longer-term Treasury rates move together in the long run due to arbitrage and yield curve dynamics. If cointegrated, a linear combination of GS3M and GS6M would be stationary, implying a stable long-term relationship. The similar forecast trends (both downward) and close last observed values (3.57 vs. 3.56) support this possibility. To confirm, an Engle-Granger or Johansen test would be needed.

Forecasts and Trends

- **GS3M**:
 - Last value: 3.57, average forecast: 3.50, downward trend (-0.07).
 - Forecast values (next 5 periods): 3.56, 3.55, 3.49, 3.46, 3.49.
 - 95% CI at period 30: [-2.05, 9.01], indicating high uncertainty in long-term forecasts.
- **GS6M**:
 - Last value: 3.56, average forecast: 3.49, downward trend (-0.07).
 - Forecast values (next 5 periods): 3.58, 3.47, 3.54, 3.46, 3.51.
 - 95% CI at period 30: [-2.23, 9.20], similarly wide, reflecting stochastic uncertainty.

Both series exhibit a slight downward trend, suggesting declining Treasury Bill rates over the forecast horizon. The wide confidence intervals reflect the stochastic nature of the series and the potential for large deviations, especially in non-stationary or near-unit-root processes. The similar forecast paths hint at cointegration, as both rates move in tandem.

Comparative Analysis

- Both series show non-stationary behavior (GS3M more clearly than GS6M) and require differencing to model effectively.
- The identical ARIMA(3,1,2) orders suggest similar underlying stochastic processes, possibly driven by common macroeconomic factors (e.g., monetary policy, inflation expectations).
- The downward forecast trends align with economic expectations if interest rates are anticipated to decline, and the close forecast values (3.50 vs. 3.49) reinforce the likelihood of cointegration.

Own interpretation

The GS3M and GS6M Treasury Bill rate series reveal a fascinating blend of stochastic behavior and economic interdependence, capturing the unpredictable yet structured nature of short-term financial markets. Their ARIMA models and non-stationary characteristics confirmed for GS3M and nuanced for GS6M highlight a random walk-like process where past rates and shocks shape future movements, reflecting the influence of monetary policy and market dynamics. The slight downward forecast trends suggest expectations of declining rates, possibly signaling anticipated economic cooling, while the wide confidence intervals underscore the inherent uncertainty in these volatile series.

The potential cointegration between GS3M and GS6M is particularly striking, as their near-identical values and parallel trajectories hint at a deeper, long-run equilibrium driven by yield curve dynamics or arbitrage. This interconnectedness, coupled with their shared stochastic structure, paints a picture of two instruments dancing to the same economic rhythm, yet each with its own subtle variations. To me, this analysis not only showcases the complexity of modeling financial time series but also underscores how Treasury rates serve as a window into the broader economic narrative, balancing predictability with the ever-present specter of uncertainty.

Answers to questions

a. Plot the two time series in the same diagram. What do you see?

The "Treasury Bill Series" plot shows GS3M and GS6M over time. Visually, both series exhibit similar patterns, moving closely together with noticeable peaks and troughs around the same periods. GS6M tends to be slightly higher than GS3M, reflecting the typical yield curve where longer maturities have higher rates than shorter ones. The parallel movement suggests a potential long-term relationship, hinting at cointegration, though with some short-term divergences. The overall trend appears non-stationary, with both series showing persistent fluctuations and no clear mean-reverting behavior.

b. Do a formal unit root analysis to find out if these time series are stationary.

The interpretation provides results for unit root tests:

- **GS3M:**
 - ADF Test: p-value = 0.5140 (> 0.05), fails to reject the null of a unit root, suggesting non-stationarity.
 - KPSS Test: p-value = 0.0100 (< 0.05), rejects the null of stationarity, confirming non-stationarity.
 - Conclusion: GS3M is non-stationary, likely $I(1)$, as first differencing ($d=1$ in ARIMA) is required for stationarity, also supported by the "Differenced Treasury Bill Series" plot where the differenced GS3M fluctuates around zero.
- **GS6M:**
 - ADF Test: p-value = 0.0032 (< 0.05), rejects the null of a unit root, suggesting stationarity.
 - KPSS Test: p-value = 0.0100 (< 0.05), rejects the null of stationarity, indicating non-stationarity.
 - Conclusion: The conflicting results suggest GS6M is borderline. The KPSS test's indication of non-stationarity aligns with the ARIMA model's $d=1$, and the differenced series in the plot looks stationary, so GS6M is likely $I(1)$ as well.

The "Differenced Treasury Bill Series" plot shows both series after first differencing, with values oscillating around zero and no clear trend, supporting that both are $I(1)$ and become stationary after differencing.

c. Are the two time series cointegrated? How do you know? Show the necessary calculations.

The interpretation suggests that GS3M and GS6M are likely cointegrated, as both are I and economic theory implies a long-term relationship. The "Treasury Bill Series with Forecasts" plot shows their forecasts moving closely together, with average forecasts of 3.50 and 3.49, respectively, supporting a stable relationship.

To formally test cointegration, we can use the Engle-Granger two-step method.

- **Step 1: Estimate the cointegrating relationship.** Assume a linear relationship. Using the last observed values, and assuming a simple regression over the series, we might estimate. However, without raw data, we hypothesize.
- **Step 2: Test the residuals for stationarity.** Compute residuals. If cointegrated, these residuals should be $I(0)$. The "Treasury Bill Series" plot shows GS3M and GS6M moving closely, so their difference residuals likely fluctuates around zero. We'd run an ADF test. Since the interpretation notes their similar trends and values, and the residuals' plot isn't provided, we infer from economic theory and visual evidence that the residuals are likely stationary, suggesting cointegration.

Thus, GS3M and GS6M are likely cointegrated, supported by their parallel movements and economic relationship.

d. **d. What is the economic meaning of cointegration in the present context? If the two series are not cointegrated, what are the economic implications?**

- **Cointegration:** In this context, cointegration between GS3M and GS6M means that the 3-month and 6-month Treasury Bill rates share a stable long-term relationship despite short-term fluctuations. Economically, this reflects the yield curve dynamics where short-term rates are linked through market expectations, arbitrage, and monetary policy effects. If cointegrated, deviations between the two rates are temporary, and they revert to a long-run equilibrium, ensuring consistency in the term structure of interest rates.
- **If not cointegrated:** If GS3M and GS6M are not cointegrated, there's no long-term equilibrium tying them together. This would imply that the rates can drift apart indefinitely, potentially signaling inefficiencies in the Treasury market, such as impaired arbitrage or disconnected monetary policy impacts across maturities. Economically, this could lead to unpredictable yield spreads, complicating investment decisions and signaling potential structural issues in the bond market or economic policy transmission.

e. **If you want to estimate a VAR model, say, with four lags of each variable, do you have to use the first differences of the two series or can you do the analysis in levels of the two series? Justify your answer.**

Since both GS3M and GS6M are $I(1)$ (non-stationary but become stationary after first differencing) and likely cointegrated, we need to consider the appropriate form for a VAR model:

- **If cointegrated:** When two series are cointegrated, a Vector Error Correction Model is typically used instead of a VAR in levels. A VECM accounts for the long-run cointegrating relationship while modeling short-run dynamics. However, if we insist on a VAR, we can estimate it in levels because the cointegrating relationship ensures the system is stable, and the VAR in levels implicitly captures the long-run relationship. So, a VAR with four lags in levels is permissible but less efficient than a VECM.
- **Stationarity requirement:** A standard VAR assumes stationarity of the variables. Since GS3M and GS6M are $I(1)$, a VAR in levels risks spurious regression unless cointegration exists. If we ignore cointegration, the safer approach is to use first differences (which are $I(0)$, as seen in the "Differenced Treasury Bill Series" plot) to ensure stationarity and avoid misleading results.